

Lifestyle & Health: Disease Risk

Answer sheet · 37 marks total

Section A — Communicable & non-communicable disease

A1. A disease caused by a pathogen (1) that can be spread from person to person (1).

A2. A disease that cannot be spread between people (1), caused by genetic/lifestyle/environmental factors (1); e.g. cardiovascular disease / type 2 diabetes / cancer / chronic lung disease (1).

Section B — Risk factors

B1. Any three: smoking (1); poor diet (1); excess alcohol (1); lack of exercise (1); obesity (1).

B2. Any two: lung disease (1); lung cancer (1); cardiovascular disease (1).

B3. This only shows a correlation between the two things (1); it does not show a mechanism/cause (1); another linked factor could explain both (e.g. those who eat more processed food may also exercise less) (1).

Section C — Cardiovascular disease & treatment

C1. Fatty deposits (atheroma) build up in the artery walls (1), narrowing them and restricting blood flow to the heart muscle (1).

C2. A stent (mesh tube) is inserted into the narrowed artery (1) and holds it open, restoring blood flow (1).

C3. Statins reduce blood cholesterol level (1), slowing the rate at which fatty deposits build up in the arteries (1).

C4. Any two points each, up to 4: heart transplant — can fully restore normal heart function (1) but carries major surgical risk and donor hearts are scarce/there may be a long wait (1); artificial heart — can be fitted more quickly / used as a bridge to transplant (1) but is mechanical and may wear out, fail, or increase risk of blood clots (1).

Section D — Homeostasis

D1. The maintenance of a constant internal environment (1), despite changes inside or outside the body (1).

D2. Any two: body temperature (1); blood glucose concentration (1); water content (1).

D3. A receptor detects a change away from the normal level (1); a coordination system (nerves or hormones) processes this (1); an effector responds to bring the level back to normal (1).

Section E — Apply it

E1. Statins: advantage — no surgery needed (1); disadvantage — must be taken long-term and can cause side effects (1). Stent: advantage — restores blood flow directly/quickly (1); disadvantage — involves a surgical procedure with associated risks (1).

E2. Each of these is a separate risk factor for cardiovascular disease (1); having multiple risk factors at once increases the overall probability of developing the disease (1), because each one adds an additional way the heart/blood vessels can be damaged, e.g. narrowed arteries plus increased strain from obesity (1).

Marking note: accept any scientifically correct alternative answer with the same reasoning, even if different from the example shown — award marks for a correctly linked point, not for matching these exact words.